

AI Engineering Master Curriculum: The Full 48-Week Path

A Complete Week-by-Week Breakdown Tailored to Split-Shift Study Windows (13:00-14:30 Weekdays / 09:00-11:30 Weekends)

Weekly Time Allocation Recap

DAYS	TIME SLOT	HOURS	WEEKLY FRAMEWORK DIRECTIVE
Mon, Tue, Thu, Fri	13:00 – 14:30	6.0 Hours	30m core theory reading / 60m practical local script building.
Wednesday	—	0.0 Hours	Absolute Rest Day. Cognitive wind-down during the mid-day split.
Saturday & Sunday	09:00 – 11:30	5.0 Hours	Deep Work. Code comprehensive frameworks, run models, finalize milestones.

Complete 48-Week Granular Syllabus

WEEK	CURRICULUM FOCUS & SUBTOPICS	WEEKLY EXECUTION TARGET & CODING DRILLS
PHASE 1: THE SOFTWARE FOUNDATIONS (MONTHS 1-2)		
Week 1	Python Syntax, Variables & Basic Lists Python Crash Course (Ch. 1-4)	Configure workspace (VS Code + GitHub Git client). Code manipulations on data primitives, slices, list comprehension loops, and basic file commits.
Week 2	Control Flow, Dictionaries & User Input Python Crash Course (Ch. 5-7)	Build programmatic logic with nested if-elif statements, complex Key-Value mapping structures, state trackers using while loops.
Week 3	Functions, Modules & Class Foundations Python Crash Course (Ch. 8-9)	Refactor repetitive scripts into functional parameter-driven modules. Build classes showcasing standard Object-Oriented Principles (OOP).
Week 4	Files, Exception Tracking & Testing Python Crash Course (Ch. 10-11)	Write dynamic read/write loops for file streams with full JSON persistence. Write explicit unit tests utilizing Python's native unittest library.
Week 5	Introduction to Algorithms & Divide & Conquer Grokking Algorithms (Ch. 1-4)	Implement Binary Search algorithms. Quantify worst-case complexity scales using Big O. Script basic recursive operations and QuickSort modules.
Week 6	Hash Tables, Graph Abstractions & BFS Grokking Algorithms (Ch. 5-6)	Map explicit network configurations using adjacency hash tables. Code a standalone Breadth-First Search (BFS) graph pathfinder from memory.
Week 7	Dijkstra's Algorithm & Path Minimization Grokking Algorithms (Ch. 7)	Build directed graph matrix schemas mapping edge weights. Code Dijkstra's shortest-path calculations handling weight updates across node pools.
Week 8	Greedy Algorithms, Dynamic Programming & KNN Grokking Algorithms (Ch. 8-11)	Write approximation sets solving the classic knapsack/covering dilemma. Build simple K-Nearest Neighbors grouping indices on grid vectors.
PHASE 2: THE MATHEMATICS & CLASSICAL MACHINE LEARNING (MONTHS 3-6)		
Week 9	Linear Algebra I: Vector Vector-Spaces & Matrices Mathematics for ML (Ch. 2)	Perform vector transformations and matrix multiplications manually on paper, then replicate programmatically via raw NumPy array functions.
Week 10	Linear Algebra II: Eigendecomposition & SVD Mathematics for ML (Ch. 4)	Compute eigenvalues and eigenvectors. Build an isolated script performing Principal Component Analysis (PCA) decomposition transformations.
Week 11		

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	Vector Calculus: Gradients & Partial Derivatives Mathematics for ML (Ch. 5)	Map out Jacobians for multi-variable inputs. Trace computational graphs to conceptually verify how errors propagate through network structures.
Week 12	Probability & Distributions: Bayes' Theorem Mathematics for ML (Ch. 6)	Write scripts calculating Gaussian curves. Compute prior, likelihood, and posterior probability bounds to create a rule-based classifier model.
Week 13	Continuous Optimization & Cost Functions Mathematics for ML (Ch. 7)	Code standard Gradient Descent parameter update loops manually in Python using mathematical partial derivative outputs.
Week 14	Math Integration: Multi-Variable ML Pipeline Mathematics for ML (Synthesis)	Combine optimization, probability, and linear systems to write a native Linear Regression engine using closed-form normal equations.
Week 15	The ML Lifecycle & Preprocessing Systems Hands-On Machine Learning (Ch. 1-2)	Construct reusable Scikit-Learn data processing workflows using ColumnTransformers, missing value imputers, and standard encoders.
Week 16	Classification Models & Evaluation Paradigms Hands-On Machine Learning (Ch. 3)	Train multi-class classifiers. Map out precision-recall trade-offs by coding custom confusion tables and ROC curve generators.
Week 17	Linear Models, Cost Paths & Regularization Hands-On Machine Learning (Ch. 4)	Implement Ridge, Lasso, and ElasticNet regressions. Track model performance trends across learning curves to spot variance errors early.
Week 18	Support Vector Machines (SVM) & Kernels Hands-On Machine Learning (Ch. 5)	Build linear and non-linear RBF kernel boundary classifiers. Tune hyperparameter parameters (C, gamma) on cross-validation sets.
Week 19	Decision Trees & Splitting Realities Hands-On Machine Learning (Ch. 6)	Train multi-level decision trees. Extract and chart feature importance ranks, and restrict growth parameters to stop overfitting.
Week 20	Ensemble Learning & Random Forests Hands-On Machine Learning (Ch. 7)	Assemble custom Voting, Bagging, Boosting (AdaBoost, Gradient Boosting, XGBoost), and Stacked model pipelines on tabular target datasets.
Week 21	Dimensionality Reduction & Clustering Layers Hands-On Machine Learning (Ch. 8-9)	Reduce messy raw vectors down using incremental PCA configurations. Set up unsupervised data groups using K-Means and DBSCAN.
Week 22	Neural Network Foundations with Keras Hands-On Machine Learning (Ch. 10-11)	Construct basic Multi-Layer Perceptrons (MLPs). Incorporate batch normalization, dropout, and custom optimization parameters into training loops.

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Week 23	Custom Models & Advanced Training Patterns Hands-On Machine Learning (Ch. 12-13)	Write custom loss functions and custom evaluation layers. Serve structured data batches smoothly via high-throughput tf.data data streaming pipelines.
Week 24	Mid-Course Milestone Project Execution Hands-On Machine Learning (Milestone)	Classical ML Capstone: Take a raw, messy dataset from Kaggle. Clean it, extract features, benchmark multiple models, tune hyperparameters, and push a detailed dashboard to GitHub.
PHASE 3: DEEP LEARNING & LARGE LANGUAGE MODELS (MONTHS 7-9)		
Week 25	PyTorch Paradigm & Tensor Frameworks Deep Learning with PyTorch (Ch. 1-3)	Create PyTorch tensors on GPU runtimes. Track autograd gradient tracking graphs. Build tensor data structures for unstructured matrix storage.
Week 26	Parametric Modeling & Gradient Optimizers Deep Learning with PyTorch (Ch. 4-5)	Write custom model loops using standard linear functions, explicit weight tracking matrices, and torch.optim optimizer blocks.
Week 27	Neural Networks for Unstructured Image Targets Deep Learning with PyTorch (Ch. 6-7)	Inherit torch.nn.Module to construct fully connected networks. Build structural transforms handling multi-channel pixels via dataset loaders.
Week 28	Convolutional Neural Networks (CNNs) Deep Learning with PyTorch (Ch. 8)	Build modular networks using conv2d, maxpool2d, and linear dropout blocks. Train an image classifier model from scratch.
Week 29	Advanced Computer Vision Models Deep Learning with PyTorch (Ch. 9-11)	Download pre-trained vision models (ResNet). Freeze initial backbone weights, swap the final output layers, and perform targeted transfer learning.
Week 30	Model Compilation & Deployment Exports Deep Learning with PyTorch (Ch. 12-14)	Save model state dictionaries. Compile trained model tracks down to optimized target script artifacts using TorchScript serialization tools.
Week 31	Text Preprocessing & Tokenizer Construction Build an LLM (From Scratch) (Ch. 1-2)	Write raw regex parsing blocks. Build a native Byte-Pair Encoding (BPE) tokenizer from scratch to map text strings into integer token sequences.
Week 32	Attention Mechanics & Multi-Head Drivers Build an LLM (From Scratch) (Ch. 3)	Write the matrix multiplications for Query, Key, and Value weights. Build causal attention mask logic to implement Multi-Head Attention blocks.
Week 33	GPT Architecture Blueprint Construction Build an LLM (From Scratch) (Ch. 4)	Assemble your GPT block framework. Connect layer normalizations, attention modules, gelu activation layers, and feed-forward networks into a unified class.

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Week 34	Pre-Training Engines & Loss Estimation Build an LLM (From Scratch) (Ch. 5)	Write an un-supervised cross-entropy loss tracking evaluation system. Set up a text generation loop using temperature scale sampling.
Week 35	Instruction Fine-Tuning Paradigms Build an LLM (From Scratch) (Ch. 6)	Format instruction-response datasets into token sequences. Fine-tune your custom model weights on dedicated cloud GPU runtimes.
Week 36	Deep Learning Integration Capstone Build an LLM (From Scratch) (Ch. 7)	LLM Milestone Project: Take your custom-built transformer, load pre-trained open weights, set up an inference script, and push the entire architecture to GitHub.
PHASE 4: PRODUCTION ENGINEERING & ENTERPRISE ARCHITECTURE (MONTHS 10-12)		
Week 37	Enterprise APIs & Orchestration Layouts AI Engineering Foundations	Build programmatic API connectors using OpenAI and Anthropic clients. Implement retry logic and fallback parameters to handle rate limits.
Week 38	Vector Embedding Pipelines & Indexing Structures AI Engineering (Vector Stores)	Process long data payloads using text splitters. Generate numerical vector arrays and index them locally within vector stores like Chroma or Pinecone.
Week 39	Retrieval-Augmented Generation (RAG) Engines AI Engineering (RAG Foundations)	Build a modular RAG pipeline. Extract relevant vector matches from data stores and feed them into prompt templates to answer user questions.
Week 40	Advanced Agents & Memory Stores AI Engineering (Agent Workflows)	Build an autonomous agent workflow using LangGraph or LlamaIndex. Equip the agent with access to multiple tools and local sql state memory databases.
Week 41	Evaluation & Cost Metrics Tracker Systems AI Engineering (Evaluation)	Implement evaluation tests (Ragas framework) to track context recall accuracy, answer faithfulness metrics, and monitor token usage costs.
Week 42	Enterprise Production RAG Application AI Engineering (Milestone)	Enterprise RAG Capstone: Build a production-ready application that securely indexes a local document library. Build a web UI using Streamlit and structure backend routes using FastAPI.
Week 43	System Architecture Planning & High-Throughput Ingestion Designing ML Systems (Ch. 1-3)	Map out data pipelines for processing both real-time streams and batch data payloads. Structure system architectures to optimize for low feature storage latency.
Week 44		

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	Feature Engineering Warehouses & Static Stores Designing ML Systems (Ch. 4-5)	Build localized feature storage structures using Redis or PostgreSQL. Write optimized data processing scripts using pandas/polars.
Week 45	Model Serving Engines & Layer Endpoints Designing ML Systems (Ch. 6-7)	Build asynchronous FastAPI inference servers. Write efficient batch prediction endpoints and implement model prediction caching structures.
Week 46	Data Drift, Concept Drift & Model Monitoring Designing ML Systems (Ch. 8-9)	Write calculation workers tracking deviations in data distributions (KS Test). Build operational alerts for target production drift patterns.
Week 47	Containerization, Orchestration & Hardware Constraints Designing ML Systems (Ch. 10-11)	Write custom multi-stage Dockerfiles to containerize your FastAPI apps. Run localized network scaling checks via docker-compose setups.
Week 48	Final Graduation Infrastructure Integration Designing ML Systems (Final Portfolio)	Grand Capstone Project: Package one of your deep learning models into a Docker container. Deploy it via a FastAPI endpoint, add vector search capabilities, and set up live logging dashboards.